

Financial Market Integrity: Detecting “Wash Trading” in NFT and Crypto Markets

Project Proposal for the Machine Learning Course

A graph-based fraud detection framework for identifying circular self-trading patterns, suspicious wallet communities, and market manipulation in blockchain ecosystems.

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Proposal Summary. This project addresses *wash trading* in NFT and crypto ecosystems, where the same actor or a coordinated group of wallets repeatedly buys and sells assets among themselves to create fake demand, inflate price, and manipulate perceived market activity. We propose a machine learning pipeline that models wallets and transactions as a graph, uses structural and temporal signals to identify suspicious communities, and classifies trading behavior as manipulative or organic.

Problem Motivation and Background

Digital assets such as NFTs and cryptocurrencies are traded in highly dynamic markets where transparency of transactions exists, but the intent behind those transactions is often hidden. One of the most damaging forms of manipulation in these markets is **wash trading**, where the same entity, or a tightly coordinated set of entities, trades with itself to create an illusion of liquidity, popularity, or price growth. Such manipulation can mislead investors, distort market statistics, and undermine trust in decentralized financial ecosystems.

A simple price prediction model does not solve this problem, because wash trading is not merely a forecasting issue; it is a *behavioral and structural fraud detection problem*. The difficulty lies in uncovering hidden relationships between wallets, repeated circular flows of value, synchronized timing patterns, and suspiciously concentrated activity around specific assets. This makes the problem both challenging and highly suitable for machine learning.

The proposed project aims to build a system that detects suspicious trading communities by combining **graph representation**, **unsupervised discovery**, and **supervised classification**. The final outcome will be a practical fraud-screening framework that is much more meaningful and technically rich than a standard regression-based market prediction task.

Project Objective

The central objective of the project is to design a machine learning framework for identifying whether a set of NFT or crypto transactions is likely to be **organic** or **manipulative**. More specifically, the

project will:

- model blockchain activity as a wallet–transaction graph,
- discover suspicious wallet clusters that exchange value in closed or repetitive loops,
- engineer meaningful structural, transactional, and temporal features,
- train a classifier to score suspicious clusters or wallets, and
- provide interpretable indicators of why a pattern is flagged as possible wash trading.

This project satisfies the course requirement of being **non-trivial** because it involves graph construction, feature engineering, clustering, classification, noisy labeling, and fraud-oriented reasoning over large real-world data.

Dataset and Inputs

We plan to use public blockchain transaction records from **Ethereum** and/or **Polygon**, obtained through publicly accessible transaction logs such as BigQuery public datasets or equivalent indexed blockchain sources. For NFT-specific context, marketplace metadata from sources such as the **OpenSea API** may be used to obtain information about collections, token identifiers, traits, and sale history.

The raw data will include wallet addresses, sender and receiver pairs, timestamps, asset identifiers, sale values, frequency of transfers, and repeated interactions between the same accounts. From this data, we can build a directed graph where:

- each **node** represents a wallet or account,
- each **edge** represents a transfer or trade,
- edge attributes may include value, token ID, collection, time gap, and number of repeated trades.

This formulation naturally captures the relational structure of the problem, which is difficult to represent using only flat tabular features.

Proposed Methodology

The proposed workflow has three major stages.

1. Graph Construction and Feature Engineering

We will first convert raw transaction logs into a transaction graph. On top of this graph, we will compute useful signals such as repeated buyer–seller interactions, reciprocity, short-cycle frequency, edge density within a small group, average holding time of an asset, concentration of trade volume, price jumps without broader market support, and ratio of trading volume to number of unique users.

These features are motivated by the intuition that wash trading often creates *artificially dense, repetitive, and low-diversity trading patterns*. A normal market usually involves broader participation and more varied interaction structure.

2. Unsupervised Discovery of Suspicious Communities

Before building a final classifier, we will apply unsupervised learning to identify suspicious local structures. Algorithms such as **DBSCAN**, **K-Means** on derived feature embeddings, or graph-community methods can be used to detect small groups of wallets that trade mostly within themselves. These groups may exhibit closed loops, unusual concentration of value flow, and repeated transactions over short intervals. This stage is important because fraud labels in blockchain data are limited and noisy. Unsupervised discovery helps narrow down candidate communities that deserve further analysis.

3. Supervised Fraud Classification

Once suspicious communities are extracted, we will assign proxy labels such as *Manipulative* or *Organic* using rule-based criteria and domain knowledge. Example signals include very high volume with very few unique wallets, repeated back-and-forth trading of the same asset, extremely short holding periods, or self-reinforcing circular transaction paths.

Using these features and cluster-level statistics, we will train a supervised classifier such as Logistic Regression, Random Forest, XGBoost, or a neural approach if time permits. The output will be a fraud likelihood score rather than only a hard binary label. This makes the system more useful for screening and further manual investigation.